

~~CS 222: Computer Organization and ARCHITECTURE~~
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~~MSE Quiz #1~~ Max Marks: 40, Max Time: 120 minutes

(Answer precisely and mention explicitly all the steps)

1. Convert -124.0625, a decimal number, to a 32-bit hexadecimal representation of its IEEE single-precision floating point (FP) equivalent. [3]
2. Compute the SUM of the following IEEE single-precision FP numbers: 3FC00000, 3EA00000 [5]
3. Compute the PRODUCT of the following IEEE single-precision FP numbers and express the result as a decimal number: BFD00000, BFB00000 [6]
4. Consider a sequential forward cache design (the "usual" design) on a processor. The system you are building uses 64-bit words for floating point control computations. You have determined the system will have the following characteristics: 300 MHz CPU; 50 MHz bus speed (6 CPU clocks = 1 bus clock). Cache has two 64-bit words per block. Bus uses burst transfer mode: 4 bus clocks for first datum, and 1 bus clock per datum in burst. Bus width is 16 data bits (16 bits transferred per bus clock during a burst). Cache hits take one CPU clock. 6% data cache miss ratio. Considering only data accesses (ignoring instruction accesses), what is the effective memory access time for this system in *nanoseconds*? [10]
5. Write Booth's Algorithm for multiplication of two integers. Explain the operation of this algorithm (-6×3). Justify its advantage in compared to the simple algorithm for multiplication discussed in class. [3+3+2=8]
6. Consider a machine with byte addressable main memory of 65536 bytes and block sizes of 8-bytes. Assume that a two-way set associative mapped cache of 256 bytes is used. Divide the memory address into the fields (Tag, set, word). Explain what happens by increasing/ decreasing number of sets in a set associative memory. [4]
7. Consider a CPU with ISA follows 4 address instruction format (4-bit opcode and 6-bit address). Write the corresponding set of instructions to solve the expression $A = (B + C) / [(B + C) * D]$. How much memory do you need to store the complete program. [4]

*****GOOD LUCK*****